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Centrifugal heat dissipation fan

Abstract

A centrifugal heat dissipation fan including a housing and a blade wheel disposed in the housing is provided. The housing has two inlets on a rotation axial direction of the blade wheel and at least one outlet on a rotation radial direction of the blade wheel. The blade wheel has a hub and a plurality of metal blades disposed and surrounding to the hub. Each of the metal blades has a body, an upper bending plate and a lower bending plate, wherein the upper bending plate and the lower bending plate are movable relative to the body. An including angle of the upper bending plate relative to the body, and an including angle of the lower bending plate relative to the body are changeable along with the rotating speed of the blade wheel.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Taiwan application serial no. 113107571, filed on Mar. 1, 2024. The entirety of each of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The disclosure relates to a centrifugal heat dissipation fan.

Description of Related Art

(3) With the development of technology, portable electronic devices, such as laptops and smartphones, have been frequently used in daily life. At the same time, in order to meet people's demand for small size and high performance, in addition to improving the above-mentioned goals of electronic devices, how to deal with the heat energy generated during the operation of electronic devices is a major issue in improving the operating performance of electronic devices. Therefore, electronic devices are usually equipped with a heat dissipation module or heat dissipation element, such as a cooling fan, to help dissipate the heat generated during operation of the electronic device

to the outside of the electronic device.
(4) The principle of the centrifugal heat dissipation fan is to rotate the blades to generate a pressure difference, so that the air from the external environment enters the fan cavity along the axial direction. Then, it is driven by the rotation of the blades and is discharged from the fan cavity along the radial direction. Generally, the cross-sectional height of blades along the rotation axial direction is positively related to the air inlet volume of the fan. That is to say, the larger the size of the blades along the rotation axial direction, the more inlet air flow is expected.
(5) However, due to manufacturing and assembly tolerances, the upper and lower edges of the blades along the rotation axial direction still need to maintain a certain distance from the inner wall of the fan housing to avoid the blades from hitting the inner wall due to deflection when rotating. Especially when the rotating speed of the blades is low, the energy generated by the electromagnet driving the blades is low at this time, so the blades are in an unstable state, and the above-mentioned deflection is prone to occur.
(6) Based on the above, how to overcome the deflection situation and increase the air inlet volume of the blades is actually a topic that relevant technical personnel need to think about and solve.
SUMMARY
(7) The present disclosure provides a centrifugal heat dissipation fan, which uses movable bending plates of metal blades to change the height of the metal blades along the rotation axial direction with the rotating speed.
(8) The present invention provides a centrifugal heat dissipation fan including a housing and a blade wheel. The housing has a pair of inlets and at least one outlet. The blade wheel is rotatably disposed in the housing. The inlets are located on a rotation axial direction of the blade wheel, the outlet is located on a rotation radial direction of the blade wheel. The blade wheel has a hub and a plurality of metal blades disposed and surrounding to the hub. Each of the metal blades has a body, an upper bending plate and a lower bending plate extending from opposite sides of the body, and the upper bending plate and the lower bending plate respectively correspond to the corresponding inlets. The upper bending plate and the lower bending plate are movable relative to the body, and an including angle of the upper bending plate relative to the body and an including angle of the lower bending plate relative to the body are changed along with a rotating speed of the blade wheel respectively.
(9) Based on the above, the centrifugal heat dissipation fan forms a body on its metal blades and an upper bending plate and a lower bending plate extending from the body. And the upper bending plate corresponds to one of the inlets in the rotation axial direction, and the lower bending plate corresponds to the other inlet in the rotation axial direction. More importantly, the upper bending plate and the lower bending plate are respectively in a movable and flexible state relative to the body, so that the including angle of the upper bending plate relative to the body and the including angle of the lower bending plate relative to the body can be changed with the rotating speed of the blade wheel.
(10) In this way, by appropriately adjusting the deformation capabilities of the upper bending plate and the lower bending plate of the metal blades, the upper bending plate and the lower bending plate will be opened relative to the body due to the increased wind force when the blade wheel is at high speed, thereby improving the ability of the metal blades to capture air volume. At the same time, the electromagnet driving the blade wheel also generates a large amount of energy due to high rotating speed requirements, thereby reducing the possibility of the blade wheel deflecting. Therefore, the aforementioned opening of the upper bending plate and the lower bending plate can be performed smoothly. On the contrary, when the blade wheel is at a low rotating speed, the upper bending plate and the lower bending plate return to their initial states of being folded relative to the body. In this way, the blade wheel deflection problem caused by the low electromagnet energy can be avoided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic diagram of a centrifugal heat dissipation fan according to an embodiment of the present invention.

(2) FIG. **2** is an exploded view of the centrifugal heat dissipation fan of FIG. **1**.

(3) FIG. **3** is a schematic diagram of metal blades of the centrifugal heat dissipation fan.

(4) FIG. **4** is a partial schematic diagram of the metal blades of FIG. **3** in different states.

(5) FIG. **5** is a top view of the blade wheel of the centrifugal heat dissipation fan of FIG. **2**.

(6) FIG. **6** is a schematic diagram of metal blades according to another embodiment of the present invention.

(7) FIG. **7** is a partial cross-sectional view of FIG. **6**.

DESCRIPTION OF THE EMBODIMENTS

(8) FIG. **1** is a schematic diagram of a centrifugal heat dissipation fan according to an embodiment of the present invention. FIG. **2** is an exploded view of the centrifugal heat dissipation fan of FIG. **1**. FIG. **3** is a schematic diagram of metal blades of the centrifugal heat dissipation fan. Referring to FIG. **1** to FIG. **3** at the same time, in the embodiment, the centrifugal heat dissipation fan **100** is used in portable electronic devices such as notebook computers or tablet computers, and includes a housing **120** and a blade wheel **110**. The housing **120** has a pair of inlets E**1**, E**2** and at least one outlet (the embodiment takes an outlet E**3** as an example). The blade wheel **110** is rotatably disposed in the housing **120**, which usually uses a drive motor as a driving force source. And the magnetic force generated by the electromagnet installed in the stator of the motor becomes the power source of the rotor equipped with the magnet. The motor is, for example, a three-phase motor.

(9) The housing **120** includes a base **122** and a cover plate **121**, wherein the base **122** has the inlet E**2** and the cover plate **121** has the inlet E**1**. And the base **122** and the cover plate **121** are combined to form the outlet E**3** and accommodate the blade wheel **110** therein.

(10) The inlets E**1**, E**2** are located on a rotation axial direction AX of the blade wheel **110**, the outlet E**3** is located on a rotation radial direction AD of the blade wheel **110**. The blade wheel **110** has a hub **111** and a plurality of metal blades **112** disposed and surrounding to the hub **111**. Each of the metal blades **112** has a body **112***c*, an upper bending plate **112***a* and a lower bending plate **112***b* extending from opposite sides of the body **112***c*. The upper bending plate **112***a* and the lower bending plate **112***b* respectively correspond to the corresponding inlets E**1**, E**2**. The upper bending plate **112***a* and the lower bending plate **112***b* are respectively in a movable and flexible state relative to the body **112***c*. The upper bending plate **112***a* and the lower bending plate **112***b* respectively change the including angle with the body **112***c* according to a rotating speed of the blade wheel **110**.

(11) FIG. **4** is a partial schematic diagram of the metal blades of FIG. **3** in different states. Referring to FIG. **3** and FIG. **4** at the same time, in the embodiment, the upper bending plate **112***a* and the lower bending plate **112***b* face the same side of the metal blades **112**, which is also equivalent to the upper bending plate **112***a* and the lower bending plate **112***b* being inclined towards the rotation direction of the blade wheel **110**. Here, the rotation direction of the blade wheel **110** is as shown in FIG. **1**. Since the thickness of the metal blades **112** in the embodiment is 0.1 mm, it has a certain degree of flexibility. Furthermore, as shown in FIG. **3**, the upper bending plate **112***a* and the lower bending plate **112***b* of the metal blades **112** are separated from the body **112***c* only by creases C**1** and C**2** respectively. Since neither the upper bending plate **112***a* nor the lower bending plate **112***b* is connected to the surrounding structure along a rotation radial direction AD, the upper bending plate **112***a* and the lower bending plate **112***b* are further provided with mobility to swing relative to the body **112***c*.

(12) In this way, as shown in FIG. **4**, the upper bending plate **112***a* and the lower bending plate **112***b* use solid line outlines to represent their undeformed initial states, and use dotted line outlines to represent their deformed states due to the blade wheel **110** reaching a high rotating speed. In other words, when the rotational speed increases and they are deformed to the dotted line outline, the including angles between the upper bending plate **112***a* and the lower bending plate **112***b* and the body **112***c* respectively increase. At the same time, it is also reflected in the size of the metal blades **112** along the rotation axial direction AX. The metal blades **112** have a length X**3** along the rotation axial direction AX before deformation, and the metal blades **112** have a length X**4** along the rotation axial direction AX after deformation, and X**4**>X**3**.
(13) Furthermore, as shown in FIG. **2**, the inlets E**1**, E**2** are limited by the structural configuration of the housing **120**, so the area of the inlet E**1** may be larger than the area of the inlet E**2**. Accordingly, the embodiment is shown in FIG. **4**. There is a minimum gap d**1** between the upper edge of the upper bending plate **112***a* and the inner top surface of the housing **120** (that is, the cover plate **121**), and there is a minimum gap d**2** between the lower edge of the lower bending plate **112***b* and the inner bottom surface of the housing **120** (that is, the base **122**), and d**1**>d**2**. In this way, the upper bending plate **112***a* and the lower bending plate **112***b* can produce different deformation effects in response to the different air inlet volume of the inlets E**1**, E**2**. In other words, when the blade wheel **110** rotates at high speed, the deformation of the upper bending plate **112***a* (or the including angle change of the upper bending plate **112***a* relative to the body **112***c*) is greater than the deformation of the lower bending plate **112***b* (or the including angle change of the lower bending plate **112***b* relative to the body **112***c*). Besides, in response to the above-mentioned difference in air inlet volume, the size of the upper bending plate **112***a* along the rotation axial direction AX is larger than the size of the lower bending plate **112***b* along the rotation axial direction AX.
(14) FIG. **5** is a top view of the blade wheel of the centrifugal heat dissipation fan of FIG. **2**. Referring to FIG. **5**, in order to achieve better air inlet efficiency and reduce the noise generated by the metal blades **112**, the optimal size of the upper bending plate **112***a* or the lower bending plate **112***b* in the rotation radial direction AD of the embodiment is 20% to 100% of the size of the metal blades **112** along the rotation radial direction AD, wherein 0% is located at the end of each metal blades **112** away from the hub **111**. In the embodiment, the size of the upper bending plate **112***a* and the lower bending plate **112***b* along the rotation radial direction AD is 32% of the size of the metal blades **112** along the rotation radial direction AD.
(15) FIG. **6** is a schematic diagram of metal blades according to another embodiment of the present invention. Referring to FIG. **3** and FIG. **6** at the same time, first, the metal blades **112** includes a section one SC**1**, a section two SC**2** and a section three SC**3** in FIG. **3**. The section one SC**1** is connected to the hub **111** (marked in FIG. **2**). The section two SC**2** is connected between the section one SC**1** and the section three SC**3**. The upper bending plate **112***a* and the lower bending plate **112***b* are located in section three SC**3**. The size X**1** of the section one SC**1** along the rotation axial direction AX is smaller than the size X**2** of the section two SC**2** along the rotation axial direction AX. And the size X**2** of the section two SC**2** along the rotation axial direction AX is smaller than the size X**3** of the section three SC**3** along the rotation axial direction AX, as shown in FIG. **3**. The metal blades **112** have gaps T**1**, T**2** between the section two SC**2** and the section three SC**3**, thus forming a step along the rotation axial direction AX. This is also one of the reasons why the upper bending plate **112***a* and the lower bending plate **112***b* can swing relative to the body **112***c*.
(16) Then, referring to FIG. **6**, in the metal blades **212** of the embodiment, which includes a section one SA**1**, a section two SA**2** and a section three SA**3**. The section one SA**1** is connected to the hub **111**. The section two SA**2** is connected between the section one SA**1** and the section three SA**3**. The upper bending plate **212***a* and the lower bending plate **212***b* are located in the section three SA**3**. The size X**4** of the section one SA**1** along the rotation axial direction AX is smaller than the size X**5** of the section two SA**2** along the rotation axial direction AX. And the size X**5** of the section two

SA**2** along the rotation axial direction AX is equal to the size X**5** of the section three SA**3** along the rotation axial direction AX.
(17) In other words, the metal blades **212** of the embodiment have shear interleaving T**3**, T**4** between the section two SA**2** and the section three SA**3** (the upper bending plate **212***a* and the lower bending plate **212***b* each intersect the section two SA**2** along the rotation direction of the blade wheel) and have no step along the rotation axial direction AX. That is, if the section two SA**2** is extended out of the virtual blade surface along the extension direction of the metal blades **212**, the orthographic projections of the upper bending plate **212***a* and the lower bending plate **212***b* on the virtual blade surface may just overlap on the virtual blade surface.
(18) The following table provides the experimentally obtained deformations of the metal blades **112**, **212** of the above two different embodiments, in which the embodiment one is the embodiment shown in FIG. **2**, and the embodiment two is the embodiment shown in FIG. **6**.
(19) TABLE-US-00001 Embodiment one Embodiment two (Blade wheel (Blade wheel diameter 38 mm) diameter 62 mm) Low Low Blade size (mm) speed Max. speed % speed Max. speed % The size of the 3.1 3.3 6.5% 2.7 3.1 14.8% metal blades along the rotation axial direction (mm)
(20) As can be seen from the above, although the area and deformation of the metal blades **112** of the embodiment one are small, the manufacturing difficulty can be reduced due to the existence of gaps T**1** and T**2**. For the embodiment two, since the metal blades **212** have a larger blade surface (and deformation), they can effectively maintain the blade surface and produce better air inlet effect (increase air inlet volume).
(21) FIG. **7** is a partial cross-sectional view of FIG. **6**. Referring to FIG. **7** and FIG. **6**, in the embodiment, the designer can increase the size X**6** of the upper bending plate **212***a* and the lower bending plate **212***b* of the metal blades **212** as much as possible according to the internal space of the housing **120** (such as the distance between the cover plate **121** and the base **122**). Herein, when the blade wheel **110** is in a stationary state that has not yet rotated, the upper bending plate **212***a* and the lower bending plate **212***b* of the embodiment conform to the aforementioned size X**5** along the rotation axial direction AX, as shown in FIG. **7**. When the upper bending plate **212***a* and the lower bending plate **212***b* are in a completely flat state relative to the section three SA**3** of the metal blades **212**, there is a size X**7**. However, due to the operating limitations of the motor, the metal blades **212** cannot technically be completely flattened. Even with sufficient rotational force, the blade structure will be restricted due to creases. In this way, the designer only needs to make the size X**7** meet the internal space requirements (such as to avoid interference with the cover plate **121** or the base **122**) of the housing **120**, and then the designer can smoothly design the required sizes of the upper bending plate **212***a* and the lower bending plate **212***b* to be larger than the size X**5** and smaller than the size X**7**.
(22) In summary, in the above-mentioned embodiments of the present invention, the centrifugal heat dissipation fan forms a body on its metal blades and an upper bending plate and a lower bending plate extending from the body. And the upper bending plate corresponds to one of the inlets in the rotation axial direction, and the lower bending plate corresponds to the other inlet in the rotation axial direction. More importantly, the upper bending plate and the lower bending plate are respectively in a movable and flexible state relative to the body, so that the including angle of the upper bending plate relative to the body and the including angle of the lower bending plate relative to the body can be changed with the rotating speed of the blade wheel.
(23) In this way, by appropriately adjusting the deformation capabilities of the upper bending plate and the lower bending plate of the metal blades, the upper bending plate and the lower bending plate will be opened relative to the body due to the increased wind force when the blade wheel is at high speed, thereby improving the ability of the metal blades to capture air volume. At the same time, the electromagnet driving the blade wheel also generates a large amount of energy due to high rotating speed requirements, thereby reducing the possibility of the blade wheel deflecting. Therefore, the aforementioned opening of the upper bending plate and the lower bending plate can

be performed smoothly. On the contrary, when the blade wheel is at a low rotating speed, the upper bending plate and the lower bending plate return to their initial states of being folded relative to the body. In this way, the blade wheel deflection problem caused by the low electromagnet energy can be avoided.

Claims

1. A centrifugal heat dissipation fan, comprising: a housing, having a pair of inlets and at least one outlet; and a blade wheel, rotatably disposed in the housing, the inlets are located on a rotation axial direction of the blade wheel, the at least one outlet is located on a rotation radial direction of the blade wheel, the blade wheel has a hub and a plurality of metal blades disposed and surrounding to the hub, each of the metal blades has a body, an upper bending plate and a lower bending plate extending from opposite sides of the body, the upper bending plate and the lower bending plate respectively correspond to the corresponding inlets, and the upper bending plate and the lower bending plate are movable relative to the body, an including angle of the upper bending plate relative to the body and an including angle of the lower bending plate relative to the body are changed along with a rotating speed of the blade wheel respectively.
2. The centrifugal heat dissipation fan according to claim 1, wherein the upper bending plate and the lower bending plate face the same side of the metal blades.
3. The centrifugal heat dissipation fan according to claim 1, wherein the upper bending plate and the lower bending plate face a rotation direction of the blade wheel.
4. The centrifugal heat dissipation fan according to claim 1, wherein the including angles between the upper bending plate and the lower bending plate and the body respectively increase as the rotating speed increases.
5. The centrifugal heat dissipation fan according to claim 1, wherein the upper edge of the upper bending plate has a minimum gap d**1** relative to the inner top surface of the housing, and the lower edge of the lower bending plate has a minimum gap d**2** relative to the inner bottom surface of the housing, and d**1**>d**2**.
6. The centrifugal heat dissipation fan according to claim 5, wherein the inlet area corresponding to the upper bending plate is larger than the inlet area corresponding to the lower bending plate.
7. The centrifugal heat dissipation fan according to claim 1, wherein the radial dimensions of the upper bending plate and the lower bending plate account for 20% to 100% of the radial dimensions of each blade, wherein 0% is located at the end of each metal blades away from the hub.
8. The centrifugal heat dissipation fan according to claim 1, wherein the thickness of each metal blades is 0.1 mm.
9. The centrifugal heat dissipation fan according to claim 1, wherein the metal blades include a section one, a section two and a section three, the section one is connected to the hub, the section two is connected between the section one and the section three, the upper bending plate and the lower bending plate are located in the section three, the size of the section one along the rotation axial direction is smaller than the size of the section two along the rotation axial direction, and the size of the section two along the rotation axial direction is smaller than the size of the section three along the rotation axial direction.
10. The centrifugal heat dissipation fan according to claim 9, wherein the metal blades have gaps between the section two and the section three and steps along the rotation axial direction.
11. The centrifugal heat dissipation fan according to claim 1, wherein the metal blades include a section one, a section two and a section three, the section one is connected to the hub, the section two is connected between the section one and the section three, the upper bending plate and the lower bending plate are located in the section three, the size of the section one along the rotation axial direction is smaller than the size of the section two along the rotation axial direction, and the size of the section two along the rotation axial direction is equal to the size of the section three

along the rotation axial direction.
12. The centrifugal heat dissipation fan according to claim 11, wherein the metal blades have shear interleaving between the section two and the section three and have no step along the rotation axial direction.